Application of Data Mining to Small Data Sets: Identification of Key Production Drivers in Heterogeneous Unconventional Resources

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Summary

In this study, we developed a data mining-based multivariate analysis (MVA) workflow to identify correlations in complex high-dimensional data sets of small size. The research was motivated by the integration analysis of geologic, geophysical, completion, and production data from a 4-square-mile study field located in the Northern Denver-Julesburg (DJ) Basin, Colorado, USA. The goal is to establish a workflow that can extract learnings from a small data set to guide the future development of surrounding acreages. In this research, we propose an MVA workflow, which is modified significantly based on the random forest algorithm and assessed using the R^2 score from K-fold cross-validation (CV). The MVA workflow performs significantly better in small data sets compared to traditional feature selection methods. This is because the MVA workflow includes (1) the selection of top-performing feature combinations at each step, (2) iterations embedded, (3) avoidance of random correlation, and (4) the summarization of each feature's occurrence at the end. When the MVA workflow was initially applied on a complex synthetic small data set that included numerical and categorical variables, linear and nonlinear relationships, relationships within independent variables, and high dimensionality, it correctly identified all correlating variables and outperformed traditional feature selection methods. Following that, a field data set consisting of the information from 23 wells was investigated using the MVA workflow aiming at identifying the key factors that affect the production performance in the study area. The MVA workflow reveals the weak correlation between production and legacy well effect. The results show that the key factors affecting production in this study area are total organic carbon (TOC) percentage, open fracture densities, clay content, and legacy well effect, which should receive significant attention when developing neighboring acreage of the DJ Basin. More importantly, this MVA method can be implemented in other basins. Considering the heterogeneity of unconventional resources, it is worthwhile to identify the key production drivers on a small scale. The outperformance of this MVA method on small data sets makes it possible to provide valuable insights for each specific acreage.

Introduction

Hydrocarbon production from unconventional reservoirs has a significant impact on the economics and geopolitics of the world. It is valuable to optimize the development of unconventional resources, but the process is complicated. Various data analytics, machine learning, and data mining approaches have been explored in an attempt to evaluate and optimize well performance. Schuetter et al. (2015) used multiple machine learning models such as random forest and support vector regression to predict the production of 476 horizontal wells in the Wolfcamp Formation. Lolon et al. (2016) studied 1,305 wells in the Middle Bakken Dolomite and 444 wells in the Three Forks Formation using multiple regression, random forest, and gradient boosting models to predict production. They concluded that water cut, total fracture fluid, and the amount of proppant pumped during completion are the most important predictors of cumulative oil production. Wang and Chen (2019) used the recursive feature elimination workflow to find the most important factors affecting the oil production of 4,790 horizontal wells in the Montney Formation. They evaluated that the six most important variables that affect production are well latitude, longitude, well true vertical depth, proppant pumped per well, well lateral length, and fluid injected per well. Rosenhagen et al. (2019) integrated geological and completion data of more than 5,000 wells in the DJ Basin to research the impact of stimulation fluid intensity on oil production using linear and polynomial regressions. Their analysis confirmed that the injection volume during stimulation shows the strongest trend with the oil production in DJ Basin.

The majority of the previous studies have available data from hundreds or thousands of wells spread over an entire basin. However, due to the heterogeneity of geology, geomechanics, and fluid properties in unconventional resources, it is critical but challenging to investigate wells within a specific smaller area. First, multiple factors can impact well performance. The data available for each well often have high dimensionality, but the sample size such as the number of investigated wells in the study area is relatively small. Second, independent variables may have nonlinear or unapparent relationships with the target parameter, which is difficult to reveal by traditional methods. The objective of this study is to design a new data mining approach to handle these difficulties and identify the hidden correlation between the well performance and controlling parameters.

The data mining method refers to a process of extracting insightful information from data sets by leveraging the methods of machine learning and statistics and presents a solution to reduce the complexity of such an analysis. Data mining has been applied in the development of unconventional resources. Grujic et al. (2015) proposed a nonparametric statistical approach to predict the production of unconventional reservoirs. To transform data sets to those with low dimensions, functional principal component analysis was applied to the decline rates of 172 horizontal wells with multiple hydraulic fracturing stages. It was found that two to three functional principal components were sufficient to describe the entire production in a shale play. However, the low-dimensional score data mixes multiple variables (also called features) and could not be used to distinguish the most important production drivers. Tahmasebi et al. (2017) combined neural networks and the genetic algorithm with the fuzzy logic approach introduced by Zadeh (1965), to identify the best combination of well

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log variables that successfully predicted critical target parameters including TOC and brittleness in unconventional reservoirs. In the study of predicting oil rate in a low-permeability reservoir (Zhang et al. 2021), the different feature importance results from principal component analysis and locality preserving projection were presented, but there was no definite conclusion about which feature can most affect the oil production.

In this study, data mining is used to identify the key features that impact production performance in a 4-square-mile area. This process is typically referred to as feature selection in the field of data science. The three types of feature selection methods commonly used are filter methods, embedded methods, and wrapper methods. The selected features are those that can result in the most accurate models and have the strongest correlations with the target. This paper will start with a brief introduction of commonly used feature selection methods, followed by a detailed description of the newly developed data mining-based MVA workflow. Then, the effectiveness of this workflow is demonstrated by the comparison with traditional feature selection methods using a complex synthetic data set. After that, this MVA method will be applied to an unconventional field data set to detect the key production drivers. Conclusions will be drawn at the end.

Methodology

In this section, we first introduce three types of commonly used feature selection methods briefly, followed by a detailed explanation of the proposed MVA workflow.

Filter methods are typically used for feature selection and are computationally inexpensive. They rank and filter data using a given metric to select the most important features. The Pearson correlation, which calculates the linear correlation (positive or negative) for each variable with the target, is most used.

Another feature selection category is the embedded methods, where specific machine learning algorithms are used. These methods are more accurate but also more computationally intensive (Mladenić 2005). The feature importance attribute embedded in the random forest algorithm is commonly used for feature selection. Random forest consists of an ensemble of decision trees, which are trained on different random subsets of the training data. During the construction of these decision trees, each node uses a single feature to split the data in a way that decreases the impurity (or variance) of the dependent variable. The mean decrease in impurity over the trees is how the algorithm ranks the features in terms of their importance.

Wrapper methods are the third primary feature selection strategy. These methods are computationally greedy and employ a search strategy that attempts to find the globally optimal feature subset based on the performance of the machine learning algorithm used. Due to their iterative nature, these methods involve the training of multiple models, which makes them less time efficient. However, wrapper methods are typically regarded as a more robust approach for feature selection. Sequential forward selection (SFS) is one of the most popular wrapper methods. It works by building a model for each feature and evaluating the performance. The top-performing feature is then evaluated in combination with every other feature and the top-performing pair is kept. Then, this pair is again evaluated with the remaining features, and this process continues until the optimal number of features is reached.

Our MVA workflow **(Fig. 1)** begins with training the random forest machine learning algorithm on each feature in the data set individually. The performance of the algorithm trained using each feature is assessed using the R^2 score from K-fold CV (default of $K = 5$). To start with the training process, we set the maximum tree depth ("max_depth") as 10 and set the minimum number of samples required to split an internal node ("min_samples_split") as 1, 2, and 4. We use the model selection function called "GridSearchCV" and the criterion of "R^2" to find the best values for the hyperparameters of "max_depth" and "min_samples_split." Then, the performance of every possible combination of feature pairs is assessed. However, since the number of possible feature combinations becomes so large when considering models with three or more features, we only select the top N (default of $N = 5$) performing feature pairs for the next step and evaluate each again using K-fold CV when combined into a trio with every combination of other features in the data set not currently in the pair. This

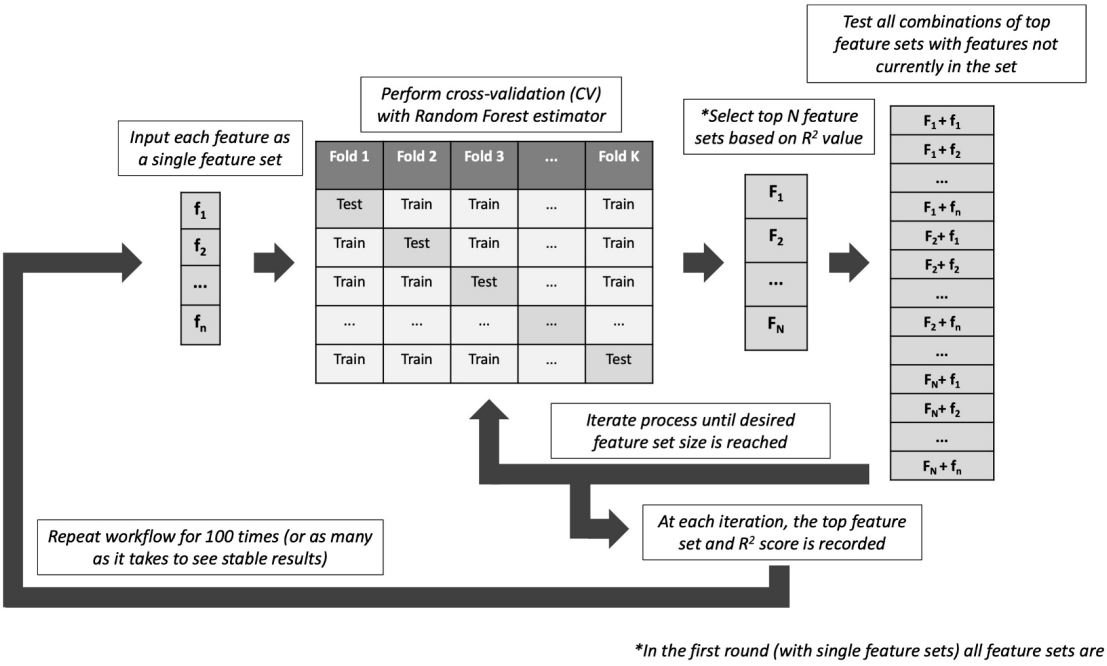

Fig. 1—Illustration of the workflow developed in this study to assess and rank independent variables in terms of their correlation with a given target. The lowercase "f" refers to a single feature, and the uppercase "F" refers to a set of one or more features.

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process of K-fold CV, selecting the top N feature subsets, and adding new features is repeated for the feature trios and for all other sizes of feature subsets until either a specified number of features or the total number of features in the data set is reached.

To ensure modeling robustness, the top N feature sets are selected in the MVA workflow, instead of only the top feature at each step like the SFS process. When using the random forest algorithm, the decision trees are trained using random subsets of the data, which may result in some amount of bias. By selecting multiple features at each step rather than just one, the impact of this potential bias is reduced. The entire workflow is iterated many times to make sure the results are robust.

The optimal number of features and the final feature importance ranking are determined through the R^2 scores recorded for each model during the workflow iterations. The optimal number of features can be read from the trend of the R^2 scores vs. the number of features included in the model. Including less than this number would result in missing information while including more would introduce unnecessary information. Among the top models utilizing only the optimal number of features, the occurrence of each feature is counted and ranked, and then the most important features can be identified. Please note that the goal of this workflow is to find the hidden correlations behind the data, but is not aimed at seeking a better prediction. Hence, the overfitting/underfitting problem will have less effect on the performance of finding important features. Additionally, because multiple models are built in the iteration process and all top-performed feature combinations are summarized and ranked at the end, the overfitting/underfitting problems are not a major concern in this study.

A step referred to as "target shuffling" is applied to evaluate the likelihood of random correlation. This step involves creating a copy of the target and randomly shuffling the order of the target values so that the only correlations present are due to random chance. The target shuffling operation maintains the statistical distribution of the original target while breaking the potential correlations. This newly created data set is called the random data set. After applying the MVA workflow to the random data set, the result is compared with that of the original, unshuffled data set. If the R^2 scores of the original data set are significantly higher than those of the shuffled data set, it means the correlations in the original data set are less likely to be random and can be captured by models.

One of the limitations of the MVA workflow is that the runtime increases exponentially with the total number of features. This challenge can be addressed by splitting up the data into smaller groups of features and running the workflow on each group. Then, the top features from each group can be selected and the workflow can be iterated using these top features. Breaking these features into groups does not have a significant impact on the final result as will be demonstrated in the synthetic example later, but it does significantly decrease the overall runtime. Additionally, running the iterations in parallel will dramatically reduce the overall runtime if multiple computational nodes are available.

Synthetic Example

To evaluate the effectiveness and reliability of the proposed MVA workflow, it was initially tested on a synthetic data set where the true correlations between variables are known.

Synthetic Example Description. The synthetic data set is generated using the pseudocode described in **Fig. 2.** It is designed to reflect the challenges of the real field data set to accurately assess the performance of the workflow. There are 60 features/variables and 60 observations. To evaluate whether the MVA workflow can detect various correlations, the designed target is determined by four variables: (1) a categorical variable (**var1**), (2) a variable with a second-degree polynomial relationship with the target (**var11**), and (3) two variables with linear relationships with the target and with each other (**var21** and **var31**). **Fig. 2** shows the crossplots of the target vs. the

Synthetic dataset consisting of 60 rows and 60 variables (**var1-var60**)

var1 = *array of random integers from 1 to 5*
var11 = *array of random numbers between -0.5 and 0.5*
var21 = *array of random numbers between -0.5 and 0.5*
var31 = *array of random numbers between -0.5 and 0.5* + 2 × **var21**

All other variables = *array of random numbers between -0.5 and 0.5*

target = **categorical response function of **var1***
+ 5 × **var11** ^{2}
+ **var21**
+ **var31**

*-0.4 where $\mathbf{var1_i} = 1$, -0.1 where $\mathbf{var1_i} = 2$, 0.5 where $\mathbf{var1_i} = 3$, 0 where $\mathbf{var1_i} = 4$, -0.3 where $\mathbf{var1_i} = 5$

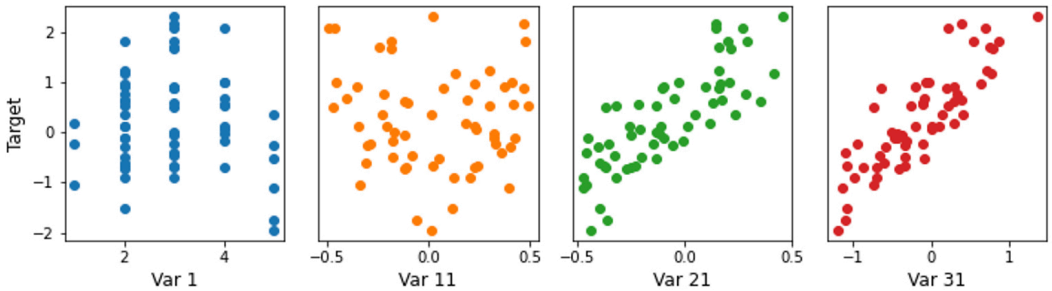

Fig. 2—Pseudocode describing the generation of the synthetic data set used to demonstrate our MVA workflow and scatter plots of the target (or dependent variable) vs. each of its correlated independent variables in the data set.

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four correlated variables. The linear correlations between the target and **var21** and **var31** are readily apparent in these plots. However, the nonlinear correlations with **var1** and **var11** are not apparent in the plots. The MVA workflow and three traditional feature selection methods are applied to this synthetic data set to see if these visually apparent and unapparent correlations can be successfully detected.

Synthetic Example Results and Discussion. Fig. 3 presents the feature ranking results of the traditional feature selection methods: (a) the Pearson correlation, (b) the random forest feature importance attribute, and (c) the SFS method using a random forest model. The Pearson correlation in **Fig. 3a** ranks **var31** as the most important feature and **var21** as the second most important. However, **var1** and **var11** are not recognized in the top ten features. This is because Pearson correlation can only identify linear relationships and **var21** and **var31** are the only variables that are linearly correlated with the target. The feature importance attribute from random forest **(Fig. 3b)** put **var21** as the most important variable followed by **var31**. However, yet again **var1** and **var11** were left out of the top ten variables. Although in theory this embedded method should be able to identify the nonlinear relationships of **var1** and **var11**, in practice, these less apparent relationships still challenge the algorithm's robustness. The SFS feature rank bar chart in **Fig. 3c** indicates that the SFS method ranked **var31, var11, var1,** and **var21** as the first, second, fourth, and fifth most important features, respectively. While the SFS method was able to identify the correlations more robustly than the filter and embedded method, it still included a feature (**var19**) that is not related to the target as the third most important variable. This is likely because a degree of randomness is involved when training the random forest algorithm. By simply selecting the top feature subset at each step, the algorithm is affected by random bias and an unrelated feature is selected over a related one.

Fig. 4 shows the results of the MVA workflow. In **Fig. 4a**, the R^2 scores of the unshuffled data are shown by the blue box plots, and the R^2 scores of the random data are shown by the red box plots. The scores of the unshuffled data are significantly higher than those of the random data, indicating that the results are statistically significant. The R^2 trend of the blue box plots shows that after only three features, the accuracy of the models plateaus, implying additional features do not contribute significantly to the target. **Fig. 4b** lists the feature importance ranking. Among all the top-performing models containing three features, 85% of these models contain **var31**, indicating

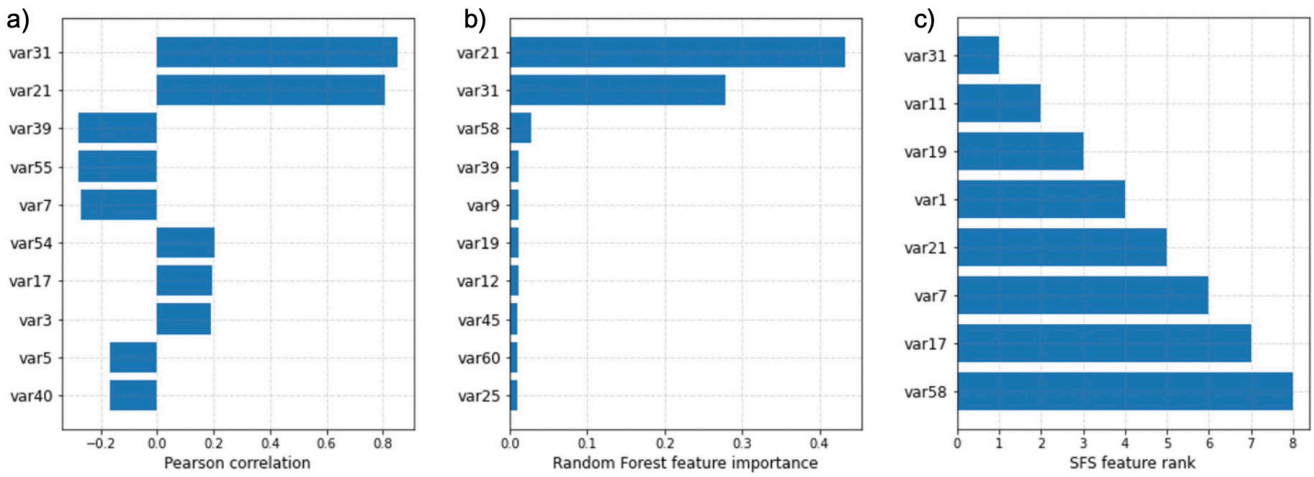

Fig. 3—Feature correlation ranking results according to (a) Pearson correlation (filter method), (b) random forest feature importance attribute (embedded method), and (c) SFS (wrapper method).

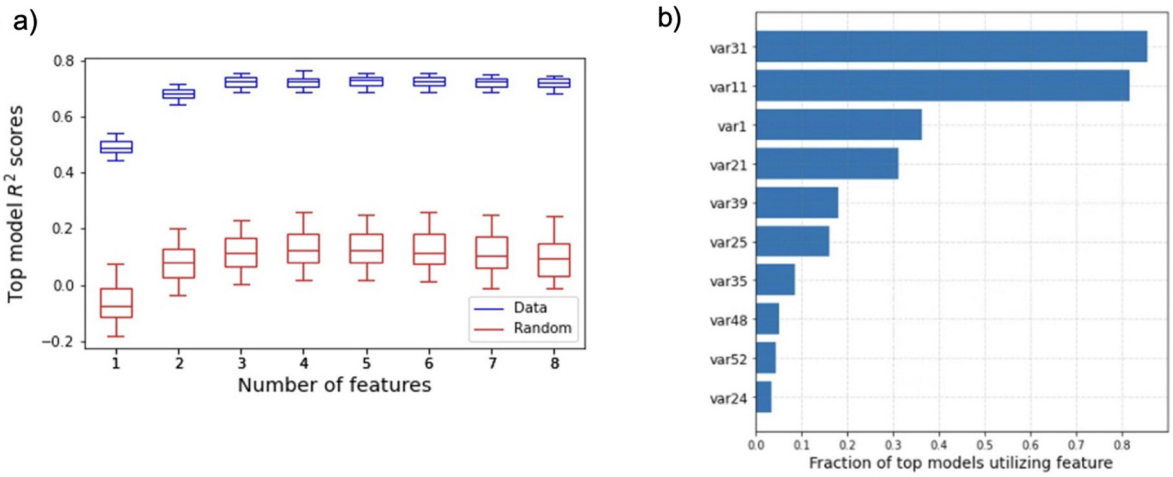

Fig. 4—Final results of the MVA workflow applied to the synthetic data set using only features that were considered the most important from each group. (a) Box plot showing the mean and standard deviation of the top R^2 scores for each of the 100 iterations of the workflow vs. the number of features included in the model for both the unshuffled (blue) and shuffled (red) data. (b) Feature importance ranking based on the fraction of the top 10% of three feature models that included each feature.

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var31 is the most important feature, subsequently followed by **var11**, **var1,** and **var21**. Comparing this result to the target synthetic data set equation in **Fig. 2**, the MVA method performs significantly better than the traditional feature selection methods by correctly correlating all four target model variables as the most important group of features.

Although this MVA workflow does not capture the exact number of features that are correlated with the target, it does capture approximately the number of features. The reason the model plateaus after only three features may be because **var21** and **var31** are correlated with each other. Therefore, including either of these two variables would by nature include some information about the other. This, in addition to the fact that **var31** has a larger variance than **var21**, could also be an explanation for why **var21** is ranked as less important than **var1** and **var11**. If they share information, including both might be viewed as redundant by the algorithm. However, the results validate that when analyzing a complex and high-dimensional data set, this MVA workflow is more reliable than the other three machine learning methods as it ranked the true correlated features above all the noncorrelated features correctly, whereas none of the other methods were successful.

Additionally, to make the synthetic data set not so idealized, we created another synthetic data set by adding 10% noise to the target. The results show that all three traditional algorithms in **Fig. 3** failed to capture the importance of the variable **var1**. However, our MVA method is still able to extract all four features (**var 31, var 21, var1, and var11**). This further demonstrates the viability of our MVA method.

Field Data Application

The MVA workflow showed promising results in the synthetic example, but the feature importance ranking in the field data may or may not be conclusive. To analyze the value of the MVA workflow on field data, the MVA methodology is used to identify the potentially important correlations in the data set, and then these highlighted correlations will be verified and interpreted using domain knowledge.

To analyze the veracity of the MVA workflow, it has been applied to a field study in the northern DJ Basin, approximately 4 miles south of the Colorado-Wyoming border. The 4-square-mile study area **(Fig. 5)** consists of 23 new horizontal wells aligned in a north-south direction that were drilled in 2019. Seven wells were targeted in the Codell Sandstone Formation while the remaining wells were drilled in the Niobrara B Chalk. The Codell well P3Bf and Niobrara well P3Df annotated in **Fig. 5a** will be discussed in more details later. The Codell Formation is located approximately 200 vertical feet below the Niobrara B Chalk. Five legacy wells, drilled in 2013 (blue solid lines), targeted the Niobrara Formation and trend in a northwest-southeast direction.

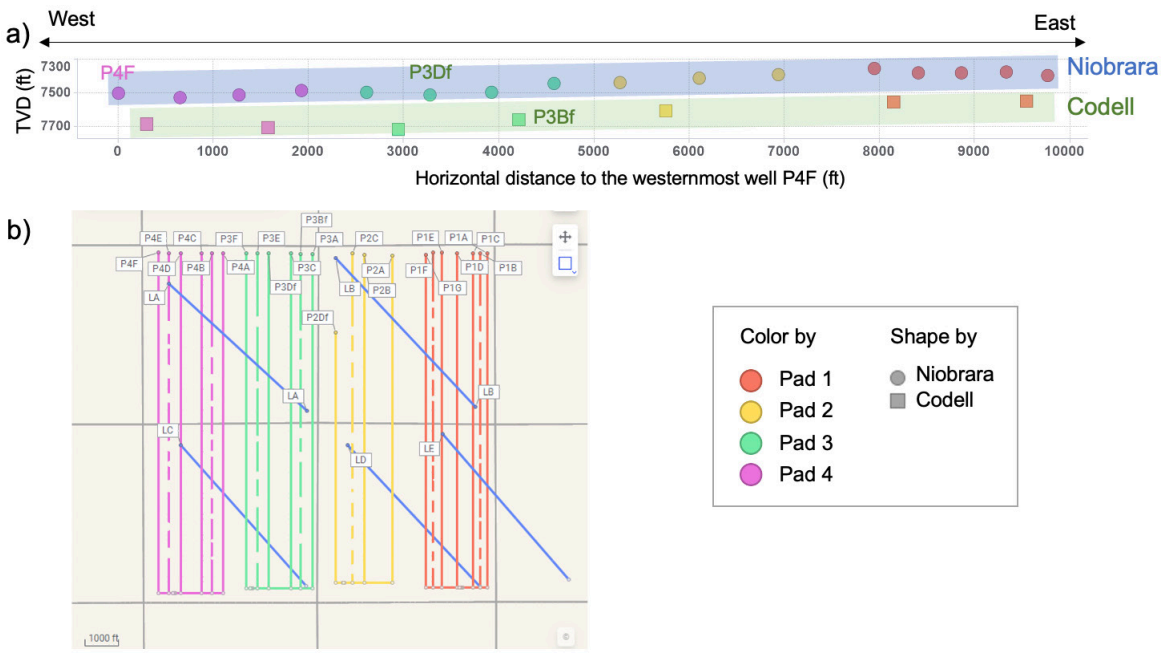

Fig. 5—(a) The gun barrel diagram shows there are 23 new horizontal wells in the study area. (b) The 4-square-mile study area consists of 23 new wells (north-south direction), which are colored by pad location. Codell wells are represented by the dashed lines whereas the Niobrara wells are shown by the solid straight lines. The five legacy wells following the northwest-southeast direction are presented by blue solid lines.

Since the target parameter is production performance, the goal of this study is to use the MVA workflow to identify the most important parameters that impact hydrocarbon production in this 4-square-mile study area. Given the 23 new wells have various lateral lengths, the production performance is represented by the 12-month barrel of oil equivalent (BOE)/ft of lateral length. Considering the production performance is highly dependent on the pressure drawdown, the pressure change for all 23 wells is investigated. It was observed that all wells have similar pressure drawdown. For instance, tubing pressure drops from ~1,400 to 200 psi within 1 month and then keeps a relatively constant value of approximately 150 psi.

For each of the 23 new horizontal wells, 21 features covering geologic characteristics and completion parameters were collected initially. The data cleaning process removed 7 features with low variance, while the remaining 14 features were used as features in the MVA workflow, as displayed in **Table 1.** These features capture the geology and completion complexity of each well. Among the 14 features, three seismic attributes [root mean square (RMS), Lambda-Rho, and variance] were extracted from a 3D seismic volume for both Niobrara B Chalk and Codell intervals and then averaged along the vicinity of each well's lateral. A dedicated case study suggests locally high seismic RMS values correspond to high TOC and low open fracture densities in the Niobrara and high clay content in the Codell;

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Features	Description
TVD (ft)	True vertical depth
Gamma	Average gamma-ray logs along each well's lateral
Seismic_RMS	Average RMS values in the vicinity of each well's lateral
Seismic_Lambda-Rho	Average Lambda-Rho values in the vicinity of each well's lateral
Seismic_Variance	Average seismic variance values in the vicinity of each well's lateral
HZspacing	Distance between one well to its nearest well in the same formation
Cross_Formation_Spacing	Distance between one well to its nearest well in the neighboring formation
LegacyWellEffect	Percentage of a new well's lateral located in the five legacy wells' drainage area
CompletionStartToEnd	Time period (days) from completion start to end date
CompletionEndToProductionStart	Time period (days) from completion end to production start date
Fluid/ft	Injected fluid amount per lateral foot (bbl/ft)
Acid_conc	Acid concentration
AvgBHPduringFrac	Average bottomhole pressure during stimulation
AvgSlurryRate (bbl/min)	Average slurry rate

Table 1—Fourteen features are used in the field data set.

Lambda-Rho is the elastic parameter related to clay percentage along the well; different seismic variance values can be caused by faults, topographic changes, or lateral heterogeneity (Downard 2021).

Fig. 6 shows the results of the field data set after 100 iterations of the MVA workflow. The results show that the R^2 scores began to decrease after five features with a mean score of 0.42. The results from the unshuffled data significantly outperformed the randomized data, indicating that there were statistically significant relationships between the target and features. Among the top 10% performance of all the five-feature models, the total occurrence of each feature was collected and ranked to find the top features. **Fig. 6b** presents that the top two features that affect production are RMS amplitude and the legacy well effect.

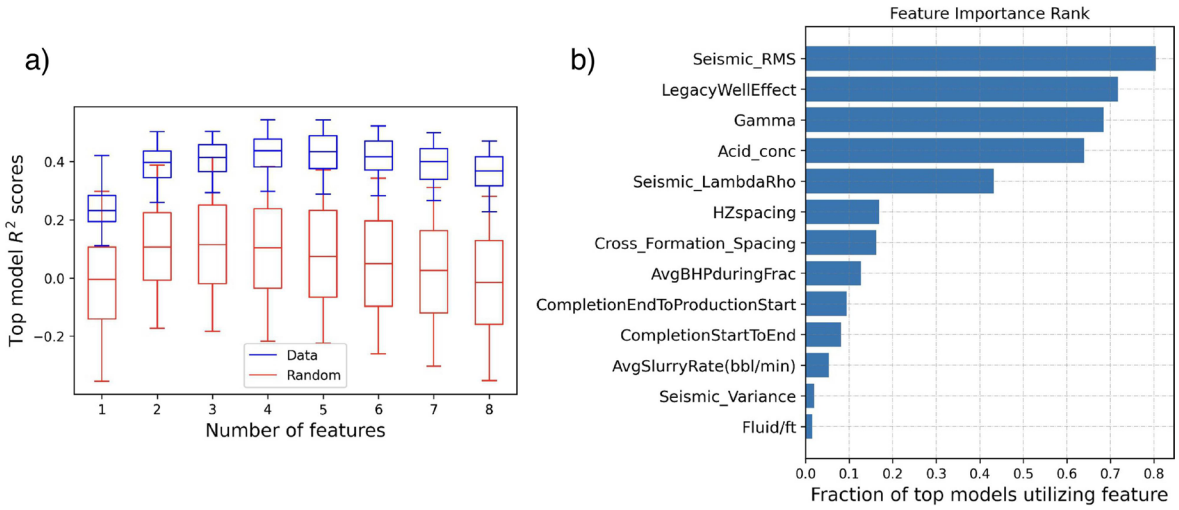

Fig. 6—Results from the MVA workflow applied to the field data. (a) The trend of the R^2 scores vs. the number of features included in the model. (b) The feature importance based on the fraction of the top 10% of models that included each feature from the models created using only five features (the number of features at which the R^2 scores began to decrease).

Seismic RMS amplitude is the top feature that correlates with production performance in this region. **Fig. 7** plots the relationship of 12-month BOE/ft with the average RMS amplitude value for each horizontal well. The 23 dots refer to the 23 new horizontal wells. Niobrara wells are marked by round dots, whereas Codell wells are by cross dots. Niobrara wells and Codell wells, by definition, are drilled within different formations and thus are investigated separately.

Downard (2021) identified two factors affecting RMS amplitude extracted at the Niobrara B Chalk interval, which are TOC percentage and open fractures, as shown in **Fig. 8.** The green and blue dots in both crossplots represent the 85 stages of the Niobrara well P3Df. **Fig. 8a** shows the relationship between RMS amplitude and TOC, indicating that RMS values along the well increase with higher TOC. **Fig. 8b** presents the negative correlation between RMS amplitude and open or partially open fractures, indicating that open fractures filled with reservoir fluid could potentially decrease seismic RMS amplitudes along the well.

Similarly, the changes in the RMS amplitude at the Codell Formation correlate with variations of shale/clay content along the wellbore, as presented in **Fig. 9.** The purple dots in the crossplot **(Fig. 9a)** refer to the 76 stages of the Codell well P3Bf.

The second most important feature impacting overall well production is tied to the legacy well effect. The legacy well effect is defined as the percentage of a new well's lateral located in the five legacy wells' drainage area. As illustrated in **Fig. 10,** the legacy well effect of Well P4A equals the sum of a and b divided by the lateral length of Well P4A, as shown in Eq. 1. The values of a and b are related to the

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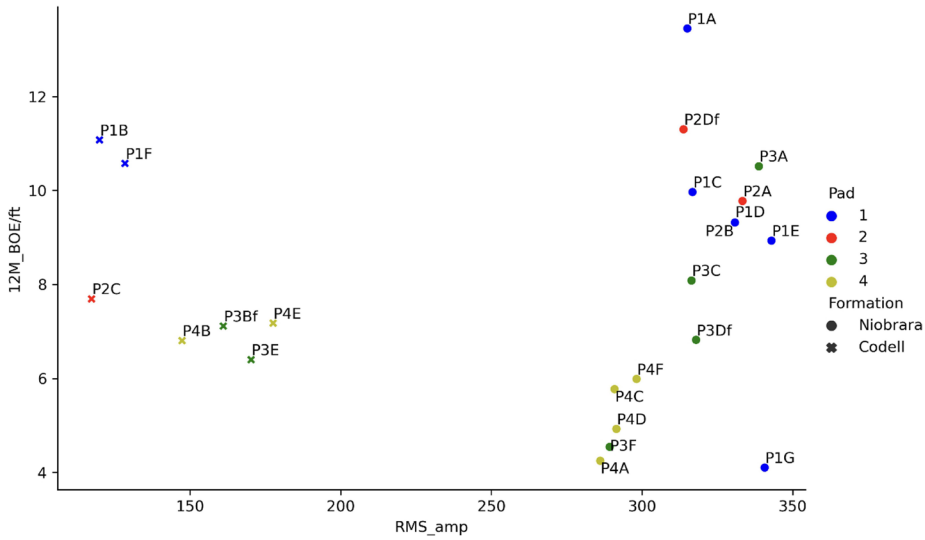

Fig. 7—Production has a positive correlation with the RMS amplitude values of Niobrara wells and has a negative correlation with the RMS amplitude values of Codell wells.

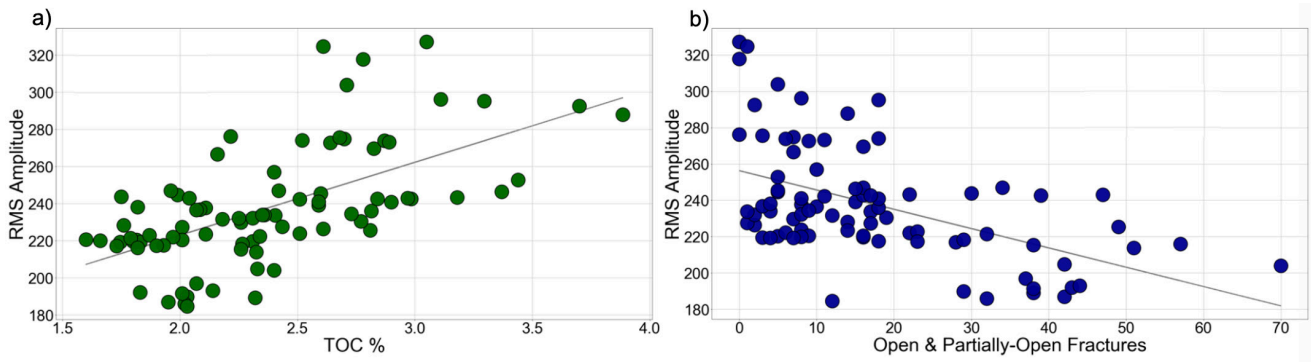

Fig. 8—Crossplots between RMS amplitude and properties along the Niobrara well P3Df. RMS is positively correlated with TOC (a) but negatively related to open and partially open fractures (b). After Downard (2021).

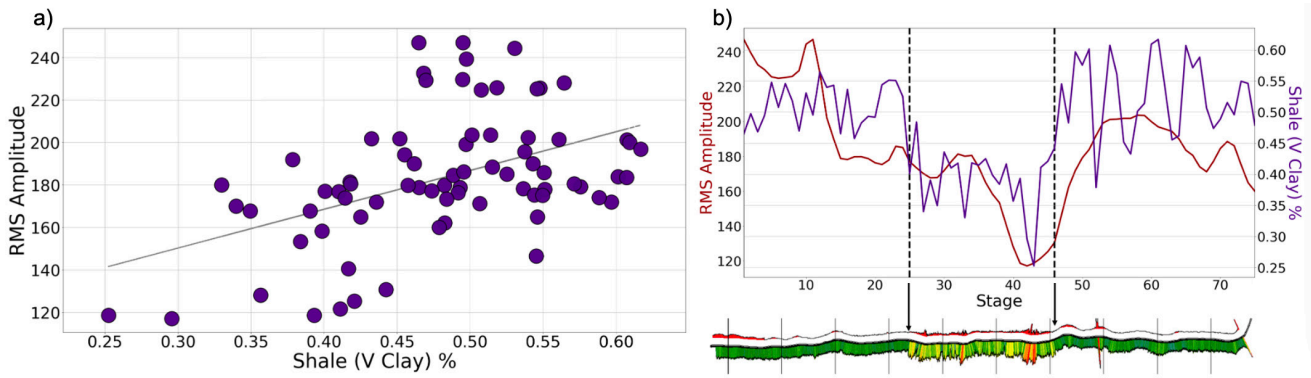

Fig. 9—Both crossplot (a) and curves (b) show that the RMS amplitude extracted along the Codell well is positively correlated to the shale/clay fraction. After Downard (2021).

half-length of the drainage area x_D (Eq. 2). Eq. 3 is used to calculate the half-length of the drainage area, where h is the reservoir thickness, ϕ is the reservoir porosity, S_w is the water saturation, B_o is the oil formation factor, and EUR is the estimated ultimate recovery. The calculated half-length of the legacy wells' drainage areas ranges from 250 to 650 ft, and the legacy well effect of new Niobrara wells is between 0 and 32%. Since all five legacy wells were drilled within the Niobrara B Chalk, Codell wells' production is not affected by the legacy wells (Schumann 2021). Ths is why we assign the legacy well effect to zero for Codell wells.

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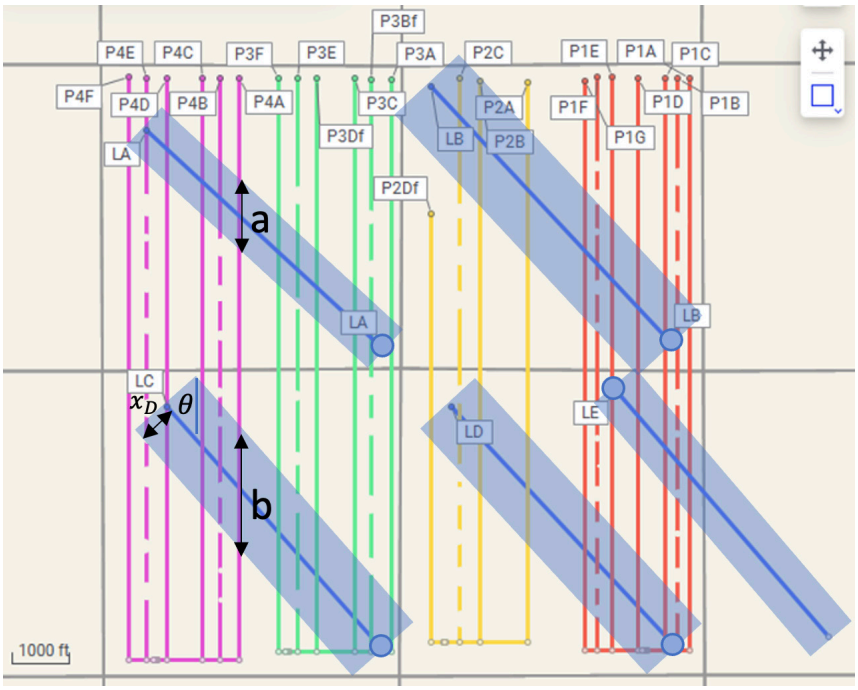

Fig. 10—Legacy wells following the northwest-southeast direction are marked as blue solid curves. Their well heads are represented by blue dots. Each legacy well's drainage area is assumed to be a rectangle. The legacy well LB has the highest cumulative oil production, which corresponds to a larger drainage area and thus its half-length is estimated to be higher than other legacy wells. The 23 new wells are drilled in an NS trend. Pads 1 and 3 are in the heel section of the legacy wells, whereas Pads 2 and 4 are in the toe section.

$$\text{Legacy well effect of Well P4A} = \frac{(a+b)}{\text{lateral length of P4A}} \tag{1}$$

$$b = \frac{2 * x_D}{\cos(\theta)} \tag{2}$$

$$x_D = \frac{\dfrac{\text{EUR}}{\text{Recovery Ratio}} * B_o}{2 * \text{Lateral length} * h * \phi * (1 - S_w)} \tag{3}$$

Fig. 11 relates the legacy well effect with the production data from the 23 new horizontal wells. The dots are colored by drilling formations and shaped by pad locations. Cross dots represent Codell wells that have no legacy well effect. Round dots refer to Niobrara wells. For the Niobrara wells, **Fig. 11** shows there is no obvious correlation between production and the legacy well effect. However, if these wells are separated by pad locations, **Fig. 12** is obtained, which shows that wells with higher legacy well effect tend to have lower production. This might be related to the pressure depletion and hydrocarbon drainage during the legacy wells' production history. It confirms that the MVA method is powerful in finding the nonapparent correlated features.

Fig. 12 only shows the relationship for the Niobrara wells on each pad. However, with Codell wells included, we can still observe similar declining trends, but with weaker correlation and smaller coefficients of determination (R^2). Because all five legacy wells are in the Niobrara Formation, we will only focus on the trends for the Niobrara wells in this research, as shown in **Fig. 12**.

In **Figs. 12a through 12d**, a negative trend between production and legacy well effect is shown for each pad, along with the values of coefficient of determination (R^2) and the slope (m). Among the four pads, the R^2 value is the lowest for Pad 4 **(Fig. 12d)**, which is mainly because P4C produces better than the wells having a similar legacy well effect. If the outlier P4C is removed (as noted by the cross symbol in **Fig. 12d**, the remaining three wells on Pad 4 fit a regression line perfectly with an R^2 value of 1, as shown by the black dashed line in **Fig. 12d**. For now, we have not identified the reason why P4C outperforms, but it is statistically acceptable to have one single outlier among the 17 data points of Niobrara wells.

Moreover, the varying slope values in **Fig. 12** show that Pads 1 and 3 have steeper slopes than Pads 2 and 4. We noticed that Pads 1 and 3 are in the heel section of the five legacy wells, while Pads 2 and 4 are in the toe section **(Fig. 10)**. One of the potential hypotheses to explain the slope difference is that the heel section of legacy wells may be more productive than the toe section. This might be caused by less effective fracture treatments in the toe due to the significant increase of friction, which lowered the treatment pressure at toe perforations and restricted the toe stages to be cleaned out effectively (Han et al. 2021). If the legacy wells' production is heel dominated, the heel section would have depleted more hydrocarbons from surrounding areas than the toe section and would more negatively affect the production of child wells nearby, which are the wells on Pads 1 and 3.

"Gamma" and "Acid_conc" are identified as the third and fourth most important features **(Fig. 6)**. The importance of "Gamma" may indicate that lithology is one main production driver for the wells in this study area. **Fig. 13a** plots the values of production and gamma for all 23 wells, which show a very weak positive correlation with a coefficient of 0.15. It is challenging to interpret the impact of gamma on production according to this figure. This might be one disadvantage of all machine learning algorithms that it presents correlation but does not determine causation.

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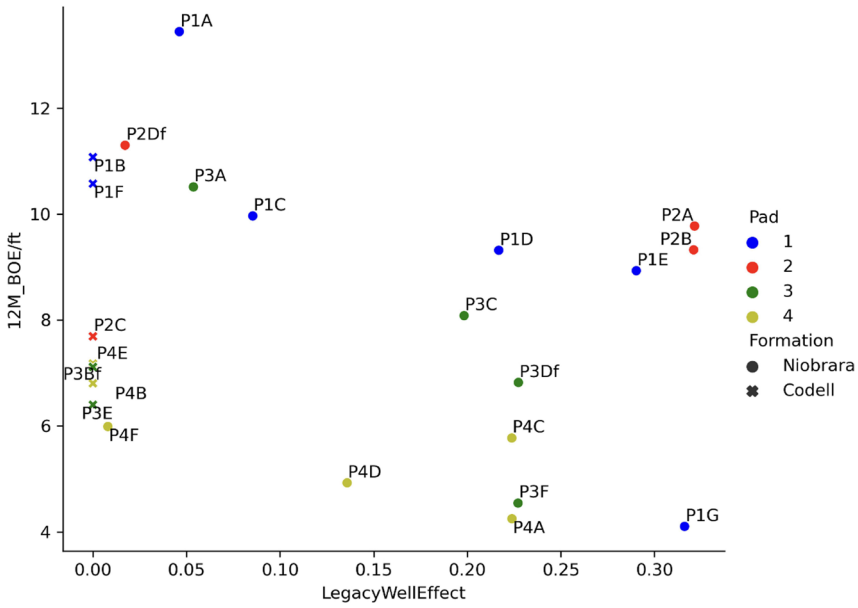

Fig. 11—Among the 23 new horizontal wells, Niobrara wells' production is negatively impacted by the legacy well effect. The trend is not clear to visualize when wells are plotted together.

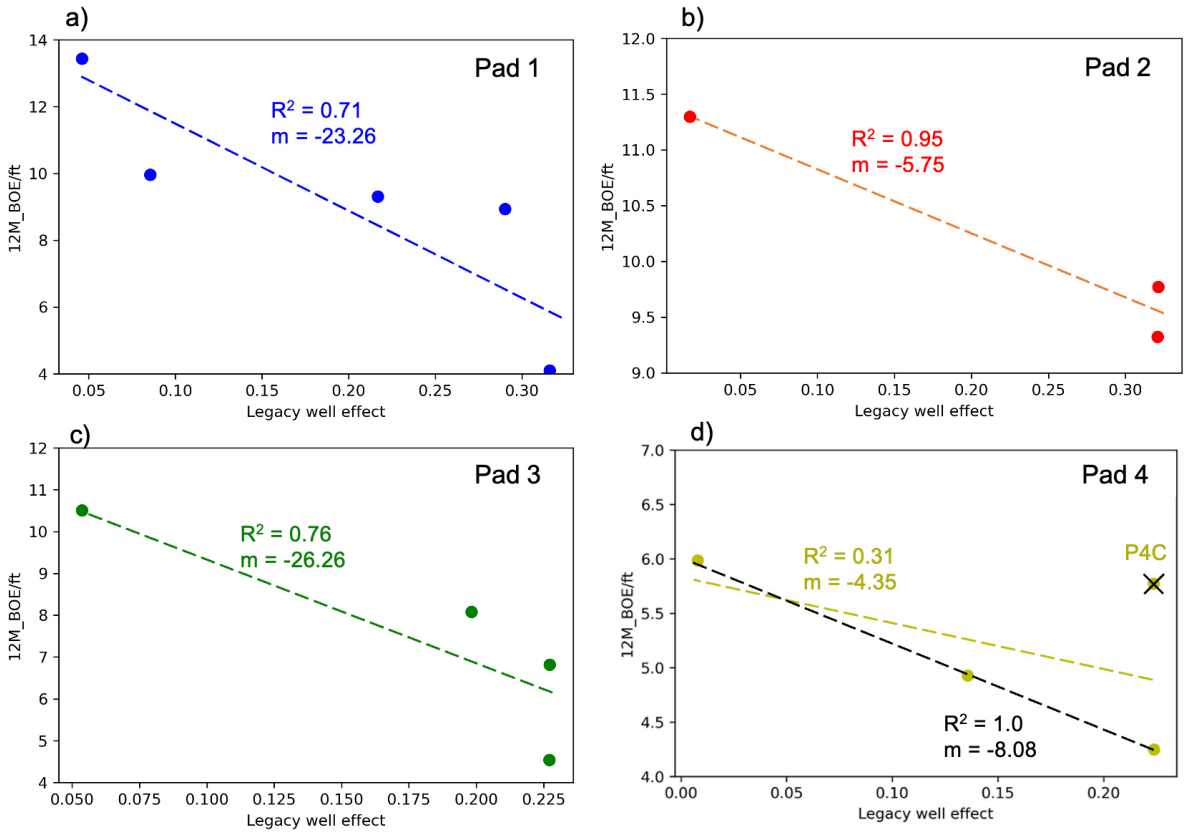

Fig. 12—There is a negative trend between production and legacy well effect when the wells are separately plotted by pad locations. The coefficient of determination (R^2) and slope *m* are annotated for the linear regression line on each plot. Pad 4 (d) has the lowest R^2 value due to the outlier P4C. When removing this data point, the remaining three wells are well matched with the black dashed line with an R^2 value of 1.0.

Fig. 13b plots the relationship between production and acid concentration. Codell wells (cross symbols) have lower acid concentrations than Niobrara wells (round dots), so acid concentration is more like a formation indicator that separates these wells by formation. No correlation is found within Codell wells, whereas a positive correlation is observed in Niobrara wells. This might be explained by the acid reaction within Niobrara Formation, which improves stimulation efficiency and leads to higher productivity.

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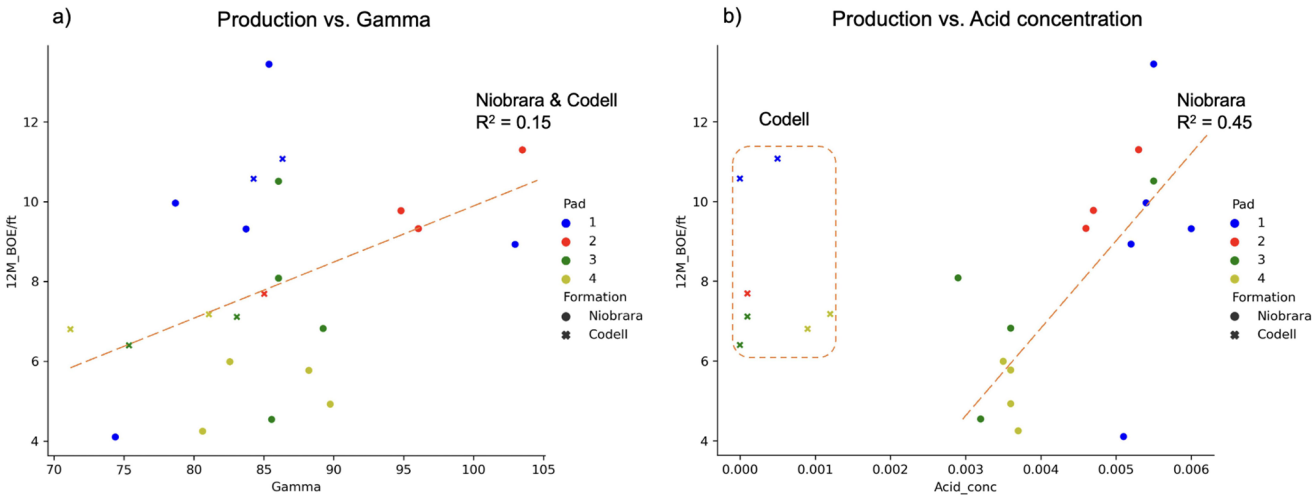

Fig. 13—(a) There is a very weak correlation between production and gamma. (b) The relationship between production and acid concentration shows that no correlation is found for Codell wells, but there is a positive trend in Niobrara wells.

Our study shows that "Fluid/ft" is one of the least important parameters affecting production **(Fig. 6)**. "Fluid/ft" here indicates the injection volume during fracturing stimulation. However, in one previous analysis based on more than 5,000 wells in the DJ Basin (Rosenhagen et al. 2019), the parameter of injection volume has been regarded as the key production driver. The difference can be explained by the size and distribution of the two data sets. As shown in **Fig. 14a**, the 23 wells have similar injection volumes, which can be separated into three groups: 30, 39, and 49 bbl/ft. The similar values allow the injection volume to be interpreted as having less impact on production. However, in **Fig. 14b**, the 5,000 wells have a wide range of injection volumes. The high variance makes it possible to capture injection volumes as the key production driver.

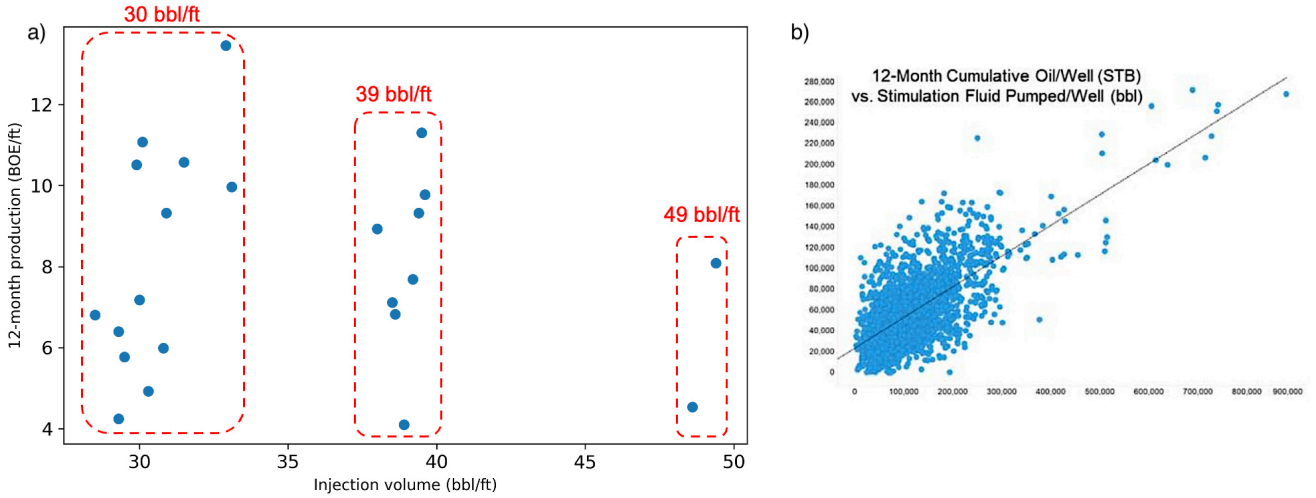

Fig. 14—(a) Production vs. injected fluid per foot for the 23 wells in our study area. (b) The 12-month cumulative oil production vs. the injection volume for more than 5,000 wells in DJ Basin.

In summary, the top four features that impact production in this study area are identified as legacy wells, RMS (TOC content, open fractures, and shale content), lithology, and acid concentration during stimulation. These top features have been widely investigated: Barhaug et al. (2022) discussed the effect of legacy wells on production performance; Cramer (2008) addressed the significance of understanding the natural fracture network; Sonnenberg (2015) concluded that geologic factors control the production in the DJ Basin; Inyang et al. (2014) discussed the importance of designing the hydraulic fracturing system to achieve higher hydrocarbon production in DJ Basin. This research is able to connect these dots and rank the importance of each variable, which provides valuable insights for operators to focus on the more important parameters.

Conclusions

This study introduced and demonstrated a new data mining-based MVA workflow to analyze complex high-dimensional small data sets. The following conclusions were drawn from this research:

- The MVA workflow outperforms three traditional feature selection methods in that all correlating variables were identified correctly as the most important features in the synthetic data set. Pearson correlation and random forest feature importance methods fail to identify the nonlinear correlation, while the SFS algorithm is unable to correctly capture the combination of top features due to random bias.

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- After applying the developed workflow in DJ Basin, it was identified that the key production drivers in this study area include legacy wells, RMS (TOC content, open fractures, and shale content), lithology, and acid concentration during stimulation. These components have been investigated by other studies, while our research is able to connect the dots and rank the importance of each variable.
- The MVA workflow indicates that a strong correlation exists between RMS and well productivity in this study area. In the Niobrara Formation, RMS amplitude is positively related to TOC percentage but negatively related to open fracture density. In the Codell Formation, RMS amplitude is positively related to clay content. These parameters should receive significant attention before designing new wells in neighboring acreage.
- The legacy well effect is identified as the second most important feature impacting production performance. The overall weak correlation between production and legacy well effect might be missed using traditional methods but was easily captured by the MVA workflow.
- This MVA method outperforms traditional methods on small data sets. Considering unconventional reservoirs are highly heterogeneous and each small acreage may need customized investigation, this MVA method can be applied to any basin or reservoir for identifying the key production drivers or other interesting aspects.

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